

Ritesh Somashekar

Python/UI Developer

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PROFESSIONAL SUMMARY

- **Python/UI Developer** with **6+ years** of experience enhancing and modernizing enterprise platforms, building **modern, scalable web user interfaces**, **distributed data-processing pipelines**, and **containerized applications** using **Python, Java, React, and FastAPI**. Proven expertise in **access control and authorization (OAuth 2.0, RBAC)**, **job queuing and scheduling**, and **workflow orchestration (Apache Airflow)**.
- Strong background in **profiling and optimizing long-running compute workloads** with **Apache Spark and PySpark**, defining **performance metrics and statistics**, and refactoring orchestration logic to improve **execution efficiency and throughput**. Experienced in **Docker, Kubernetes, and CI/CD pipelines** on **AWS and Azure**, delivering secure, **HIPAA-compliant** systems.
- Demonstrated success in **redesigning UI architecture** to improve **information hierarchy, usability, consistency, and layout** across functional modules. Adept at **technical design discussions, code reviews, and technical documentation**, collaborating effectively across engineering, product, and research teams.

TECHNICAL SKILLS

Programming & Scripting: Python (pandas, NumPy, asyncio), Java, JavaScript, TypeScript, SQL (PostgreSQL, Oracle), Bash

Web UI & Frontend: React, HTML5, CSS3, Responsive Design, Component Libraries, Design Systems, Figma, Accessibility (WCAG), Data Visualization (Plotly, Power BI, Tableau)

Backend, APIs & Security: FastAPI, Flask, RESTful Microservices, API Versioning, OAuth 2.0, JWT, Role-Based Access Control (RBAC), Authentication & Authorization

Job Scheduling & Orchestration: Apache Airflow (DAGs), Celery, Job Queuing & Prioritization, Concurrent Execution, Kafka, Redis, Event-Driven Workflows

Containerization & DevOps: Docker, Kubernetes, Jenkins, CI/CD, Git, DVC, MLflow

Distributed Processing & Performance: Apache Spark, PySpark, Hadoop, Hive, ETL Pipelines, Profiling (cProfile), Performance Metrics, Throughput Optimization

Cloud & Databases: AWS (S3, EC2, Lambda, Glue, RDS, SageMaker), Azure (Stream Analytics, Data Services), MongoDB, Snowflake, Delta Lake

ML & Analytics: PyTorch, scikit-learn, HuggingFace Transformers, BERT/BioBERT, Optuna

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA.

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India.

PROFESSIONAL EXPERIENCE

George Mason University | UI/UX Developer

Fairfax, VA | August 2024 - Present

- Modernized the **UI architecture** of a multi-user research analytics platform through a full redesign in **React** and **Python (FastAPI)**, improving **information hierarchy, usability, consistency, and layout** across functional modules (data ingestion, job monitoring, results, and administration).
- Designed and developed **modern, scalable web user interfaces** backed by a reusable **component library and design system**, applying **responsive design** and **accessibility (WCAG)** standards and serving data to the UI with **sub-200ms latency**.
- Enhanced **access control and authorization** by implementing **role-based access control (RBAC)** and token-based (JWT) authentication on FastAPI endpoints, enforcing role- and project-level permissions for datasets, jobs, and results.
- Built **job queuing and scheduling** for long-running compute jobs (NLP runs over **30K+ documents**) on the **ORC Hopper HPC cluster**, supporting **concurrent execution, priority-based dispatching, and efficient resource allocation**.
- Streamlined **end-to-end workflows** by identifying bottlenecks in data-preparation and training stages and **refactoring orchestration logic**, standardizing **15GB** of data across 6 datasets and reducing preparation time by **30%**.
- Analyzed **run-processing pipelines**, developed **performance metrics and statistics** (runtime, queue wait, throughput) with **MLflow**, and applied optimization strategies that cut iteration cycles from **3 days to under 18 hours**.

- Containerized platform services with **Docker** and automated delivery through **CI/CD pipelines**, reducing release cycles from **2 weeks to 5 days** and enabling **A/B testing** of UI and backend changes.
- Collaborated in **technical design discussions** and **code reviews** (Git, DVC), and authored **technical documentation** and **engineering best practices**, reducing onboarding time for new contributors to **under 2 days**.

Carelon Global Solutions | *Python Developer (Client: Accenture)*

India | January 2019 – July 2023

- Designed and extended **access control and authorization** for enterprise **RESTful microservices** using **OAuth 2.0**, role-based permissions, and **API versioning**, securing **HIPAA-compliant** data exchange for a U.S. healthcare client.
- Enhanced **job scheduling and queuing** in **Apache Airflow**, configuring DAG priority weights, pools, and concurrency limits to support **concurrent execution, prioritization, and efficient resource allocation**, reducing manual effort by **25%**.
- Streamlined **end-to-end workflows** by identifying bottlenecks and **refactoring orchestration and dependency logic**, improving coordination across ingestion, processing, and serving components and reducing project turnaround time by **15%**.
- **Profiled and optimized long-running distributed workloads** by migrating from **Hive to Apache Spark/PySpark** and tuning partitioning, caching, and executor resource allocation, significantly reducing processing time and increasing throughput.
- Built scalable **distributed data-processing pipelines** using **PySpark** and **AWS (S3, Glue, SageMaker)** with **Delta Lake**, improving data reliability by **30%** across ingestion and processing stages.
- Developed **performance metrics and statistics** dashboards using **Azure Stream Analytics** and **Power BI** to monitor pipeline throughput and system health, reducing unplanned downtime by **20%**.
- Containerized and orchestrated Python services using **Docker** and **Kubernetes** with **Jenkins CI/CD**, improving system uptime by **20%** and enabling real-time EHR integration via **REST APIs**.
- Developed high-throughput **Python/Java** services, including **BERT/BioBERT** clinical entity extraction (**80% accuracy, 55% faster processing**) and fault-tolerant **MongoDB** data models for unstructured data.
- Collaborated with engineering, product, and **Accenture** client teams in **technical design discussions** and **code reviews**, and contributed **technical documentation** and engineering best practices.